EPA's Air Quality Time Series Project (EQUATES) Data Use Policy Statement

- The EQUATES datasets result from extensive emissions, meteorology, and air quality modeling. The major components of this modeling effort and the point(s) of contact for each component are listed below. It is strongly recommended that careful attention be paid to the contents of the metadata file associated with these data to evaluate dataset limitations, restrictions, and intended use. For work that will significantly depend on these modeling datasets, we strongly encourage you to consult with the relevant developers to further explain the quality and limitations of the data. These types of discussions are most productive at an early stage in the work to gauge the suitability of these data for the given application and/or research effort. Where appropriate, we strongly encourage offering co-authorship to key team members on publications and conference presentations, which helps ensure continued support from the EQUATES team for that activity.
- If you are forwarding any of these datasets to a collaborator, we require that you include this Data Use Policy Statement in the forwarded material.
- When referring to the EQUATES datasets in published material, please use the following citation and DOI:
  US EPA (2021). *EQUATES CMAQv5.3.2 output (*version 1.0) [data file and documentation]. DOI: https://doi.org/10.15139/S3/F2KJSK
- When referring to the EQUATES emissions data, please using the following citation and DOI:

  Foley, K. M., Pouliot, G. A., Eyth, A., Aldridge, M. F., Allen, C., Appel, K. W., Bash, J. O., Beardsley, M., Beidler, J., Choi, D., Farkas, C., Gilliam, R. C., Godfrey, J., Henderson, B. H., Hogrefe, C., Koplitz, S. N., Mason, R., Mathur, R., Misenis, C., Possiel, N., Pye, H. O. T., Reynolds, L., Roark, M., Roberts, S., Schwede, D. B., Seltzer, K. M., Sonntag, D., Talgo, K., Toro, C., Vukovich, J., Xing, J., and Adams, E., 2023: 2002-2017 Anthropogenic Emissions Data for Air Quality Modeling over the United States. Data in Brief, https://doi.org/10.1016/j.dib.2023.109022.
- These datasets are products of the U.S. Government and are intended for public access and use. Unless otherwise specified, all data produced by the U.S EPA is, by default, in the public domain and are not subject to domestic copyright protection under 17 U.S.C. § 105. More details on the U.S. Public Domain license are available here: http://www.usa.gov/publicdomain/label/1.0/
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- Additional information on the EQUATES project is available here: www.epa.gov/cmaq/EQUATES

Topic	Developer(s)	Contact Information	Reference/Documentation
Meteorology			
WRF, MCIP simulations; meteorology trends and evaluation	Rob Gilliam	gilliam.robert@epa.gov	
Emissions			
Emissions from volatile chemical products (VCP)	Karl Seltzer Havala Pye	seltzer.karl@epa.gov pye.havala@epa.gov	https://doi.org/10.5194/acp-21-5079-2021
Emissions from fires (wildfires, prescribed, agricultural)	George Pouliot Jeff Vukovich	pouliot.george@epa.gov vukovich.jeffrey@epa.gov	

Mobile source emissions from MOVES3	EPA MOVES Team	mobile@epa.gov	https://www.epa.gov/moves/latest-version-motor-vehicle-emission-simulator-moves
NH_3 emissions from agriculture	Jesse Bash	bash.jesse@epa.gov	
Emissions from oil and gas sector	Jeff Vukovich	vukovich.jeffrey@epa.gov	
Emissions from Electric Generating Units	Caroline Farkas	farkas.caroline@epa.gov	
NO produced by lightning (estimated online in CMAQ for 12US1 domain, estimated offline for hemispheric domain)	Daiwen Kang Jeff Vukovich	kang.daiwen@epa.gov vukovich.jeffrey@epa.gov	https://doi.org/10.5194/gmd-12-3071-2019 https://doi.org/10.5194/gmd-12-4409-2019
Biogenic and soil NO emissions (estimated online in CMAQ for 12US1 domain, estimated offline for hemispheric domain)	George Pouliot Barron Henderson	pouliot.george@epa.gov henderson.barron@epa.gov	
Dust emissions (estimated online in hemispheric CMAQ simulation)	Rob Gilliam Jon Pleim	gilliam.robert@epa.gov pleim.jon@epa.gov	https://doi.org/10.1002/2016MS000823
Emissions from sources not listed above	George Pouliot Alison Eyth Jeff Vukovich	pouliot.george@epa.gov eyth.alison@epa.gov vukovich.jeffrey@epa.gov	
Boundary Conditions			
Use of hemispheric CMAQ model output for regional CMAQ simulations	Christian Hogrefe Rohit Mathur	hogrefe.christian@epa.gov mathur.rohit@epa.gov	
Air Quality Modeling with CMAQ			
Organic aerosol (aero7 version) including $PM_{2.5}$ major components	Havala Pye Ben Murphy	pye.havala@epa.gov murphy.benjamin@epa.gov	https://www.epa.gov/cmaq/how-cite-cmaq#Particle_Microphysics_and_Chemistry
Marine halogen chemistry	Golam Sarwar	sarwar.golam@epa.gov	https://doi.org/10.1021/acs.est.5b01657
Ozone trends and variability on regional-to-hemispheric scales	Rohit Mathur Christian Hogrefe	mathur.rohit@epa.gov hogrefe.christian@epa.gov	
Cloud chemistry	Kathleen Fahey	fahey.kathleen@epa.gov	https://doi.org/10.5194/gmd-10-1587-2017
Gas-phase chemistry in CB6	Bill Hutzell	hutzell.bill@epa.gov	https://doi.org/10.1016/j.atmosenv.2018.11.060
Surface exchange processes (deposition, bidirectional exchange) using Surface Tiled	Jesse Bash	bash.jesse@epa.gov	

Aerosol and Gaseous Exchange (STAGE)			
Precipitation-adjusted and bias corrected wet deposition; dry and wet deposition annual totals	Sarah Benish Kristen Foley Jesse Bash	benish.sarah@epa.gov foley.kristen@epa.gov bash.jesse@epa.gov	
Stratosphere–troposphere exchange on 3D ozone distributions (dynamic potential vorticity scaling parameterization for hemispheric CMAQ simulations)	Rohit Mathur	mathur.rohit@epa.gov	https://doi.org/10.5194/acp-16-10865-2016
General questions about model setup and evaluation (CONUS 12US1 domain)	Wyat Appel Kristen Foley	appel.wyat@epa.gov foley.kristen@epa.gov	https://doi.org/10.5194/gmd-2020-345
General questions about model setup and evaluation (hemispheric 108NHemi domain)	Christian Hogrefe Rohit Mathur Kristen Foley	hogrefe.christian@epa.gov mathur.rohit@epa.gov foley.kristen@epa.gov	